Phosphaalkenes

Gold(I) Complexation of Phosphanoxy-Substituted Phosphaalkenes for Activation-Free LAuCl Catalysis

Miki Kato, Yasuhiro Ueta, and Shigekazu Ito*[a]

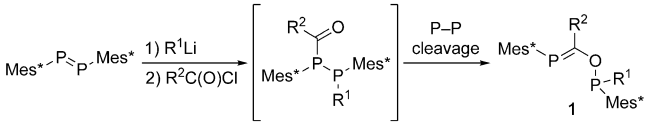
Abstract: The phosphanoxy-substituted phosphaalkene bearing the P=C–O–P skeleton can be prepared from diphosphene Mes*P=PMes* (Mes* = 2,4,6-tBu$_3$C$_6$H$_2$), and their use for catalysis is of interest. In this paper, complexation of the phosphanoxy-substituted phosphaalkenes with gold are investigated, and the catalytic activity of the mono- and bis(chlorogold) complexes are subsequently evaluated. Reaction of the P=C–O–P compound with (tht)AuCl (tht = tetrahydrothiophene) showed dominant coordination on the sp^3 phosphorus, and complete coordination on the sp^2 phosphorus required removal of tetrahydrothiophene. Atoms In Molecules (AIM) analysis based on the X-ray structure of the mono(chlorogold) complex indicated a pseudo coordinating interaction between the gold center and the P=C unit. The bis(chlorogold) complexes displayed conformational isomerism, and catalyzed the cycloisomerization/alkoxycyclization of 1,6-enyne and for hydration of terminal alkyne without activation treatment. Even the mono(chlorogold) complexes catalyzed the alkoxycyclization reactions without a silver co-catalyst, indicating that the alcohols were effective in activating the AuCl unit.

Introduction

Phosphaalkenes bearing P=C double bond(s) have been utilized as strong π-accepting ligands because of the low-lying LUMO levels comparable with that of carbon monoxide (CO). Although the phosphorus–carbon multiple bonds are normally too reactive to isolate, the kinetic stabilization method using sterically encumbered substituents enables the heavier unsaturated bonds to be protected from reactive species.[1,2] So far, many kinds of isolable low-coordinated organophosphorus compounds have been reported,[3,4] and several transition-metal-based Lewis acid catalysts have been developed.[5,6]

We have utilized kinetically stabilized phosphaalkenes to develop homogeneous gold catalysis.[7] Homogeneous gold catalysis featuring soft π-acidic character has greatly progressed over the past few decades, and many catalytic molecular transformations have been established.[8,9] In general, catalytically active gold complexes bearing ligated phosphines should not contain chloride, because the Cl$^-$ ion binds with gold strongly and prevents coordination of the substrates enough to cause the following molecular transformations.[10] However, several phosphaalkene-chlorogold(I) complexes can catalyze molecular transformations without activation using silver co-catalysts.[5a,7,11] It is plausible that the strong π-accepting character of P=C can promote the Lewis acid functionality on the gold center(s) even in the presence of Cl$^-$ ion. In addition, most of the phosphaalkene-chlorogold(I) catalysts can be recovered after the reactions, which is promising to develop sustainable chemical processes.[12]

The first isolable diphosphene Mes*P=PMes* (Mes* = 2,4,6-tBu$_3$C$_6$H$_2$, supermesityl) is one of the key compounds involved in the recent progress on the chemistry of multiple bonds including heavier main group elements.[13] The "true phosphobenzene" has been utilized as a starting material for the synthesis of low-coordinated phosphorus compounds by cleaving the P=P double bond.[3,4b] Previously, we described a metathesis-type transformation of Mes*P=PMes* by using an organolithium reagent and an acyl chloride, affording phosphanoxy-substituted phosphaalkene **1** via the P–P bond cleavage in the acyl-substituted diphosphane intermediates (Scheme 1).[14] The presence of both sp^2 and sp^3 phosphorus atoms in **1** is quite attractive as one of the non-symmetric P2 ligand systems. Whereas attempted synthesis of chelate transition-metal complexes bearing **1** has been unsuccessful until now, use of **1** for the chemistry of gold complexes has been possible.



Scheme 1. P=P bond cleavage of diphosphene Mes*P=PMes* affording phosphanoxy-substituted phosphaalkene **1**.

[a] *M. Kato, Dr. Y. Ueta, Prof. Dr. S. Ito*
Department of Applied Chemistry
School of Materials and Chemical Technology
Tokyo Institute of Technology
2-12-1-H113 Ookayama, Meguro-ku, Tokyo 1528552 (Japan)
E-mail: ito.s.ao@m.titech.ac.jp
Homepage: http://www.apc.titech.ac.jp/~sito/index-e.html
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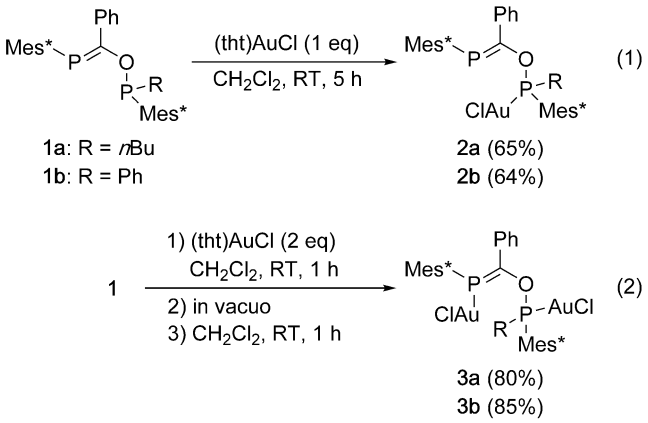
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In this paper, we describe the preparation and catalytic activity of chlorogold(I) complexes bearing **1**. Although the P=C unit in **1** basically shows inferior coordination ability to AuCl, the sp^2 phosphorus atom is effective to increase the catalytic activity of the gold center even in the presence of chloride ions. In the course of structure determination, the bis(chlorogold) complexes displayed dynamic conformational changes. Interestingly, weak interaction between the P=C group and the sp^3 phosphanoxy-coordinated gold center in the mono(chlorogold) complex might be effective to catalyze the cyclization process under the activation-free conditions accompanying alcohols.

Results and Discussion

Preparation and structure of chlorogold complexes bearing the ligated P=C–O–P system

According to our previous report,[14] phosphanoxy-substituted phosphaalkenes **1** were prepared from diphosphene Mes*P=PMes*, organolithium reagents, and benzoyl chloride. Reaction of **1** with 1 equivalent of (tht)AuCl (tht = tetrahydrothiophene) allowed predominant coordination of the gold on the sp^3 phosphorus in the phosphanoxy group, and the corresponding mono(chlorogold) complexes **2** were isolated in moderate yields (Scheme 2, Eq (1)). Shortening the reaction time (2 h) gave almost the same results. On the other hand, reaction of **1** with 2 equivalents of (tht)AuCl was sluggish and gave mixtures of bis(chlorogold) complexes **3** together with **2**. This result indicates that, in contrast to most phosphaalkenes we have investigated so far,[7] the coordination ability of the P=C unit in **1** is considerably weak, probably owing to the presence of the oxo unit, and comparable with that of tetrahydrothiophene. To avoid cumbersome purification processes in the synthesis of **3**, the volatile materials including tetrahydrothiophene were removed from the reaction mixture of **1** and 2 equivalents of (tht)AuCl, and subsequent addition of the solvent (dichloromethane) was conducted. This procedure was successful, and only bis(chlorogold) complexes **3** were isolated in good yields (Scheme 2, Eq. (2)). It should be mentioned that de-coordination of AuCl from the sp^2 phosphorus generating gold nanoparticles was somewhat problematic in the preparation and isolation of **3b**.



Scheme 2. Synthesis of chlorogold(I) complexes bearing the phosphanoxy-substituted phosphaalkenes.

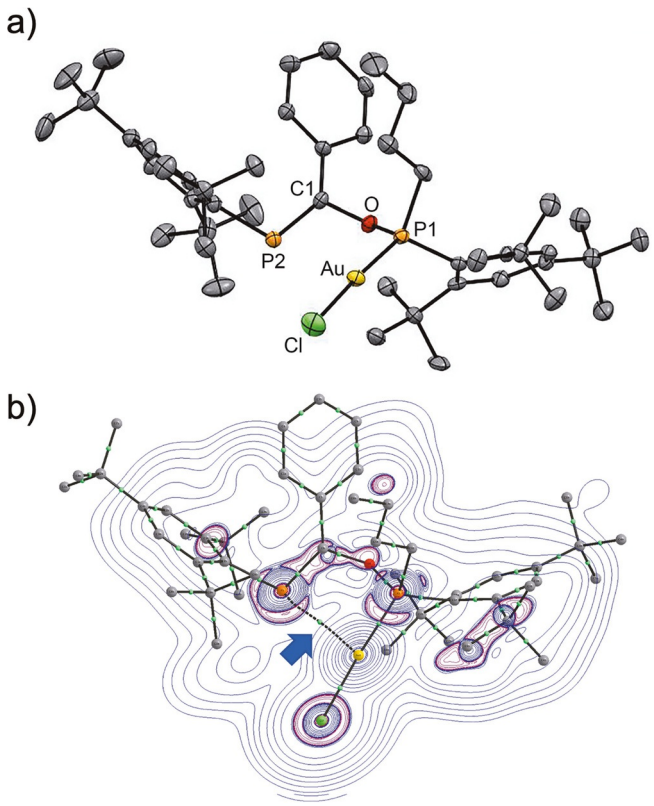


Figure 1. a) An ORTEP drawing of **2a** at the 50% probability level. Hydrogen atoms are omitted for clarity. One of the *o*-CMe_3 groups in the P1-connected Mes* unit is disordered, and the atoms of the predominant occupancy factor (0.68) are shown. Bond lengths (Å) and angles (°): Au–Cl 2.296(3), P1–Au 2.232(2), O–P1 1.633(5), O–C1 1.431(8), C1–P2 1.698(8); P1-Au-Cl 176.5(1), P1-O-C1 123.5(4). b) A 2D contour plot of the Laplacian distribution $\nabla^2\rho(r)$ in **2a** in the plane of P1, P2, and Au. The bond critical points (BCPs) are colored light green. The AIM analysis was conducted based on the single point calculation at the ωB97XD/SDD(Au), 6–31 + G(d) level. At the BCP between the sp^2 phosphorus atom and the gold center (blue arrow), $\rho(r)$ and $\nabla^2\rho(r)$ of 0.0144 e $Bohr^{-3}$ and + 0.0332 e $Bohr^{-5}$ are characterized, respectively.

Figure 1a shows an ORTEP drawing of **2a** indicating that the P=C moiety directs toward the gold center with the P···Au distance of 3.404 Å. The P2–C1 distance is almost the same as that of **1** (1.701 Å).[14] The structure of **2a** indicates the pseudo chelate interaction, which could be related to the lower coordination ability of **2** (see above). The Au–Cl distance of 2.296 Å is slightly elongated compared with **3** (see below), indicating that the P=C unit might be effective to increase the Lewis acidity on the gold center. The Atoms in Molecule (AIM) study[15,16] showed the trajectory from the sp^2 phosphorus atom and the gold center, indicating both a pseudo σ-donation from the lone pair and a back-donating π-coordination to the P=C unit (Figure 1b).[9j] Thus, the presence of the P=C–O skeleton in **2a** might increase the Lewis acidity by weakening the Au–Cl interaction. The natural bond orbital (NBO) analysis could not give scalable parameters depicting the small interaction between the gold center and the sp^2-phosphorus atom.

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Although the solution quality was not good enough to determine the metric parameters completely owing to the lack of X-ray diffraction data, the X-ray structure of **3a** (R = *n*Bu) revealed two conformers without clear Au–Au contact, and intramolecular Au···Au distances of 5.563 and 5.663 Å were observed in the crystalline state (Figure 2). The P=C–O–P skeleton in **1** is comparable with dppe [1,4-bis(diphenylphosphino)-ethane], providing the bis(chlorogold) complex without the intramolecular aurophilic interaction.[17]

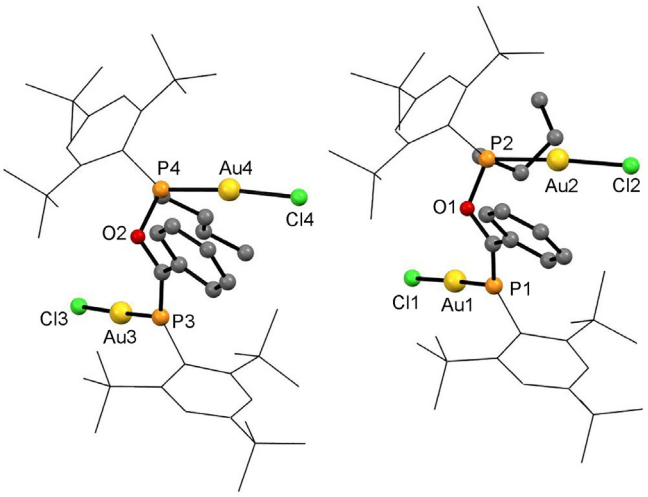


Figure 2. A drawing of the qualitatively determined structures of **3a** in the crystalline state. Hydrogen atoms are omitted for clarity. Bond lengths around the gold centers (Å): P1–Au 2.226(4), P2–Au2 2.238(3), Au1–Cl1 2.264(6), Au2–Cl2 2.285(3), P3–Au3 2.230(4), P4–Au4 2.234(3), Au3–Cl3 2.270(6), Au4–Cl4 2.278(3).

In the NMR analyses of **3b** (R = Ph), the broadened ^{31}P NMR (121 MHz) signals were observed around $\delta = 140$ and 180 ppm at 298 K (Figure 3a), which is in sharp contrast with **3a** (δ_P = 138.7, 154.0 ppm; $^3J_{PP}$ = 40.4 Hz). The ^{31}P NMR spectrum of **3b** might correlate with the rapid interconversion between at least two conformers. Neither separation nor sharpening of the ^{31}P NMR signals was observed between 183–323 K. Preliminary DFT calculations for **3b** as well as the X-ray structure of **3a** indicated that the aurophilic interaction[18] might not be advantageous. Therefore, we speculated that the presence of four aromatic rings in **3b** might produce the propeller-type conformations (Figure 3b). In the ^{1}H variable-temperature (VT)-NMR analyses (400 MHz, [D$_8$]THF), the aromatic protons showed coalesce around 243 K, and the protons of the Ph-C=P unit became non-equivalent below 213 K ($\Delta\nu \sim 360$ Hz, see the Supporting Information). By using the Gutowsky–Holm's formula,[19] the activation energy ($\Delta G^{\ddagger}$) of 11 kcal mol^{-1} was roughly estimated.

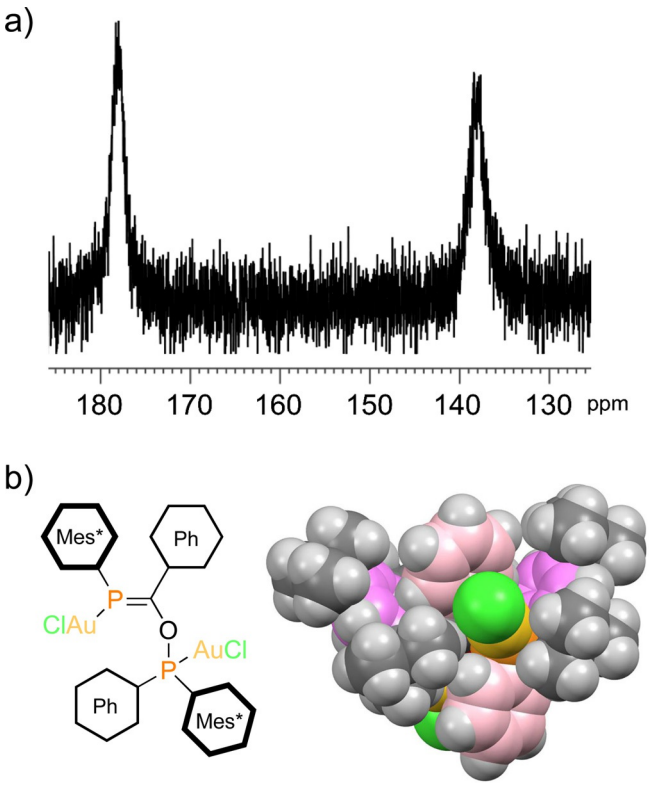
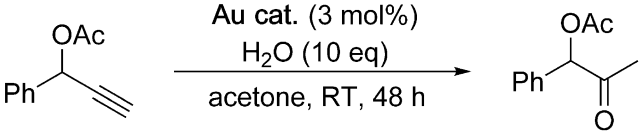


Figure 3. a) A ^{31}P{^{1}H} NMR spectrum of **3b** (121 MHz, 298 K, C_6D_6). b) A space-filling drawing of the DFT-simulated structure of **3b** at the ωB97XD/SDD(Au), 6-31G(d) level.

Catalytic activity of bis(chlorogold) complexes bearing the ligated P=C–O–P system

As we have demonstrated so far, phosphaalkene-chlorogold(I) complexes can catalyze several molecular transformations by activating the terminal alkyne units without needing a silver co-catalyst.[7a–f] Bis(chlorogold) complexes **3** can also be used for several catalytic reactions without Ag salts. Tables 1 and 2

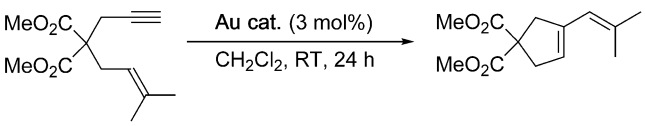
Table 1. Gold-catalyzed hydration of 2-propynyl acetate.



Au cat.	Yield [%][a]	Recovery [%][b]
3a	69	31
3b	60	40

[a] Determined by ^{1}H NMR spectroscopy.

Table 2. Gold-catalyzed cycloisomerization of 1,6-enyne.

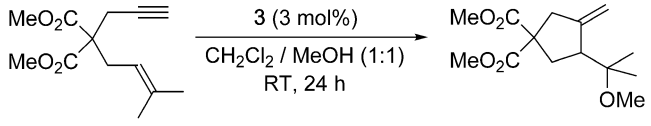


Au cat.	Yield [%][a]	Recovery [%][b]
3a	95	0
3b	93	7

[a] Determined by ^{1}H NMR spectroscopy.

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Table 3. Gold-catalyzed alkoxycyclization of 1,6-enyne.



3	Yield [%][a]	Recovery of 1,6-enyne [%][a]	Recovery of **3** [%][b]	Yield of **2** [%][b]
3a	73	23	84	16
3b	97	3	67	33

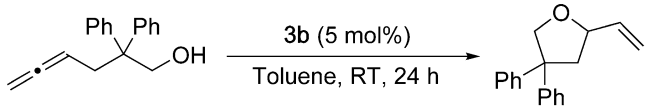
[a] Determined by ^{1}H NMR spectroscopy. [b] Determined by ^{31}P NMR spectroscopy.

display the results for the hydration of 2-propynyl acetate and cycloisomerization of 1,6-enyne catalyzed by **3**, respectively. Both **3a** and **3b** showed comparable catalytic activity with the mononuclear chlorogold complex bearing Mes*P=CPh$_2$.[7a] Complexes **3** showed no decomposition in the hydration and cycloisomerization reactions.

Table 3 summarizes the alkoxycyclization of 1,6-enyne catalyzed by **3**. Whereas **3a** showed a moderate catalytic activity, **3b** gave the methylenecyclopentene product in an excellent yield. However, both **3a** and **3b** showed de-coordination of AuCl from the P=C unit in the catalytic process, and the mononuclear complexes **2** were obtained after the alkoxycyclization reaction. Although the detailed mechanism has been unclear, the sensitive P(sp^2)–Au bond to alcohol might correlate with the lower coordination ability of the sp^2 phosphorus in **1**. A considerable amount of the de-coordination product **2b** was found after the catalytic reaction using **3b**, which correlated with the instability observed during the preparation process.

The superior catalytic activity of **3b** for the alkoxycyclization prompted us to examine other catalytic reactions including hydroalkoxylation. As shown in Table 4, 2,2-diphenylhexa-4,5-dien-1-ol was converted into the corresponding 2-vinyltetrahydrofuran in a low yield (entry 1). Use of AgSbF$_6$ as a co-catalyst improved the yield up to 91% (entry 2).

Table 4. Gold-catalyzed intramolecular hydroalkoxylation of allenyl alcohol.



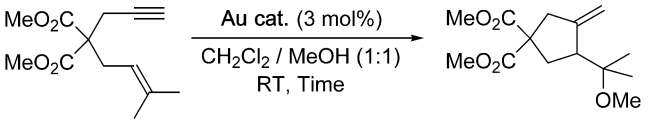
Entry	Additive	Yield [%][a]	Recovery [%][a]
1	–	11	89
2	AgSbF$_6$ (10 mol%)	91	9

[a] Determined by ^{1}H NMR spectroscopy.

Mono(chlorogold) complexes (2) for activation-free gold catalysis

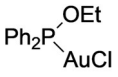
In addition to bis(chlorogold) complexes **3**, catalytic reactions using the mononuclear complexes **2** were also examined. The hydration and the cycloisomerization reactions did not occur in the presence of **2**. On the other hand, alkoxycyclization of 1,6-enyne was catalyzed in the presence of **2** (Table 5). Both **2a** and **2b** provided small amounts of the cyclization product after 24 h (entries 1, 2). Elongation of the reaction time was effective to improve the yield up to 62% with **2a** (entry 3) and, to our delight, 95% with **2b** (entry 4). No decomposition of **2** was observed after the catalytic processes. Warming at 40 °C was effective to accelerate the reaction (entries 5, 6). The ethoxydiphenylphosphine-chlorogold(I) complex **4**[20] showed no catalytic activity, indicating that the presence of the P=C unit in **2** is responsible for the gold-catalyzed the alkoxycyclization under the activation-free conditions (entry 7). The intramolecular hydroalkoxylation of allenyl alcohol was not facilitated by **2b** even in the presence of AgSbF$_6$ (see the Supporting Information).

Table 5. Alkoxycyclization of 1,6-enyne catalyzed by mononuclear complexes **2**.



Entry	Au cat.	Time [d]	Yield [%][a]	Recovery of 1,6-enyne [%][a]
1	**2a**	1	19	81
2	**2b**	1	15	85
3	**2a**	6	62	38
4	**2b**	6	95	<5
5	**2b**[b]	1	25	75
6	**2b**[b]	3	80	20
7	**4**	1	0	>99

[a] Determined by ^{1}H NMR spectroscopy. [b] Reaction was carried out at 40 °C.



4

As discussed above, the presence of the P=C-O skeleton in **2a** might increase the Lewis acidity by interacting with the gold center. Also, possible hydrogen-bond interactions between the chloride in **2** and methanol might be advantageous to the activation-free catalytic alkoxycyclization. Although a detailed analysis should be conducted, Figure 4 suggests a plausible interaction that increases the Lewis acid character on the gold center in **2**.[21,22] In the alkoxycyclization reaction with **3**,

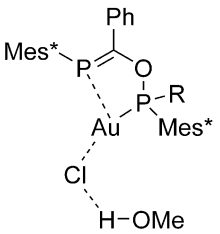
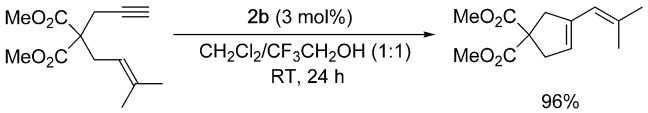


Figure 4. Plausible interaction between **2** and methanol increasing the Lewis acidity of the gold center.

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methanol might facilitate the de-coordination of AuCl from the sp^2 phosphorus atom (see Table 3). The ^{31}P NMR titration study showed that the amount of methanol affected the ^{31}P chemical shifts and the spin–spin coupling parameters, although the deviation was considerably small (see the Supporting Information).

To explore the effects of alcohol on the catalytic activity of the mono(chlorogold) complex in more detail, reaction of 1,6-enyne in the presence of other alcohols was investigated. Whereas *t*BuOH deactivated the catalytic functionality of **2b**, use of 2,2,2-trifluoroethan-1-ol (CF_3CH_2OH) was quite effective to promote the cycloisomerization process (Scheme 3). Although the low nucleophilicity of CF_3CH_2OH was useless for the alkoxylation process, the increased Lewis acid character by the perfluorinated ethanol was suitable to activate the alkyne unit leading to the vinylcyclopentene product.



Scheme 3. Cycloisomerization of 1,6-enyne promoted by **2b** in the presence of 2,2,2-trifluroethan-1-ol.

Conclusion

We described the utility of the phosphaoxy-substituted phosphaalkenes for activation-free gold catalysis. The mononuclear structure of **2** displayed weak interaction between the P=C unit and the gold center, which might increase the Lewis acidity. Bis(chlorogold) complexes **3** did not contain the aurophilic interaction, and showed conformational isomerism probably owing to the presence of four aryl substituents. Bis(chlorogold) complexes **3** containing the P(sp^2)–Au bond catalyzed cycloisomerization/alkoxycyclization of 1,6-enyne and hydration of 2-propynyl acetate without any activation treatment. Interestingly, even mono(chlorogold) complexes **2** showed catalytic activity in the alkoxycyclization, indicating the additional effects of alcohol for increasing the Lewis acidity. The findings of this study suggest a novel design for sustainable gold catalysis as well as for elucidation of catalytic mechanisms promoted by the phosphaalkene unit.

Experimental Section

All experiments were carried out under inert atmosphere (nitrogen or argon) unless otherwise noted. ^1H NMR, ^{13}C NMR, and ^{31}P NMR spectra were measured with a Bruker AVANCE 300 spectrometer or a JEOL JNM-ECZ400S (400 MHz) spectrometer. Chemical shifts of ^1H NMR were expressed in parts per million downfield from $CHCl_3$ as an internal standard ($\delta = 7.26$ ppm) in $CDCl_3$, C_6H_6 as an internal standard ($\delta = 7.16$ ppm) in C_6D_6. Chemical shifts of ^{13}C NMR were expressed in parts per million downfield from $CDCl_3$ as an internal standard ($\delta = 77.16$ ppm) in $CDCl_3$, C_6D_6 as an internal standard ($\delta = 128.4$ ppm). Chemical shifts of ^{31}P NMR were expressed in parts per million downfield from 85% H_3PO_4 as an external standard ($\delta = 0$ ppm). Mass spectra were measured with a JEOL JMS-T100LC spectrometer. X-ray diffraction data were measured with a Rigaku RAXIS-RAPID diffractometer. The structures were solved by a direct method (SHELXL-2014).[23] The X-ray structure solution and refinement were carried out by using the Yadokari-XG software.[24] Density Functional Theory (DFT) calculations were carried out by using Gaussian 09 (B.01) package.[25]

Preparation of 2a

A mixture of **1a** (71.5 mg, 0.10 mmol) and (tht)AuCl (32.1 mg, 0.10 mmol) in dichloromethane (1 mL) was stirred for 5 h at room temperature. After the volatile materials were removed in vacuo, the resultant residue was washed with *n*-hexane (5 mL) to afford **2a** as a pink solid (61.1 mg, 65%). ^1H NMR (300 MHz, $CDCl_3$): $\delta =$ 0.57 (t, 3H, $J = 6.7$ Hz), 0.85–1.20 (m, 5H), 1.33 (s, 18H), 1.46 (s, 18H), 1.50 (s, 18H), 2.30–2.50 (m, 1H), 6.22 (d, 2H, $J = 7.9$ Hz), 6.77 (pt, 2H, $J = 7.8$ Hz), 6.95 (t, 1H, $J = 7.1$ Hz), 7.31–7.35 ppm (m, 4H); $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$): $\delta = 14.1$, 23.4 (d, $J = 15.9$ Hz), 25.9, 31.0, 31.4, 32.9 (d, $J = 6.7$ Hz), 33.0 (d, $J = 6.8$ Hz), 34.6, 35.0, 37.1 (d, $J = 33.3$ Hz), 38.1 (d, $J = 8.5$ Hz), 40.0–41.0, 122.5 (d, $J = 8.4$ Hz), 124.7 (d, $J = 1.8$ Hz), 126.8 (d, $J = 6.3$ Hz), 127.2 (d, $J = 2.0$ Hz), 128.0 (d, $J = 3.4$ Hz), 131.0 (dd, $J = 61.7$ Hz, $J = 3.8$ Hz), 138.7 (d, $J =$ 3.3 Hz), 138.9 (d, $J = 3.5$ Hz), 151.3, 151.9 (d, $J = 3.3$ Hz), 155.3 (d, $J =$ 2.1 Hz), 155.7 (d, $J = 1.5$ Hz), 187.9 ppm (dd, $J = 44.0$ Hz, $J = 8.8$ Hz); $^{31}P\{^1H\}$ NMR (121 MHz, $CDCl_3$): $\delta = 123.9$ (d, $J = 93.0$ Hz), 198.7 ppm (d, $J = 93.0$ Hz); HRMS (APCI) calcd for $C_{47}H_{72}AuClNaOP_2$ $[M+Na]^+$: 969.4310; found: 969.4312.

X-ray crystallographic data of **2a**: $C_{47}H_{72}ClOP_2Au$, $M_W = 947.40$, crystal dimensions $= 0.230 \times 0.200 \times 0.080$ mm^3, orthorhombic, $Pna2_1$ (#33), $a = 20.3117(4)$, $b = 22.7508(8)$, $c = 10.2382(7)$ Å, $V =$ 4731.1(4) Å^3, $Z = 4$, $T = 123$ K, $\lambda = 0.71075$ Å, $\rho_{calc} = 1.330$ g m^{-3}, $\mu_{MoK\alpha} = 3.265$ mm^{-1}, $F_{000} = 1952$, $-26 \le h \le 26$, $-29 \le k \le 28$, $-12 \le l \le 13$, 45194 total reflections ($2\theta_{max} = 54.97^\circ$), 10327 unique ($R_{int} =$ 0.0369), $R = 0.0383$ ($\sigma > 2\sigma(I)$), 0.0441 (all data), $wR2 = 0.1146$ ($\sigma >$ $2\sigma(I)$), 0.1201 (all data), $S = 0.952$ (482 parameters). Deposition number 2025940 contains the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service www.ccdc.cam.ac.uk/structures.

Preparation of 2b

A mixture of **1b** (73.5 mg, 0.10 mmol) and (tht)AuCl (32.1 mg, 0.10 mmol) in dichloromethane (1 mL) was stirred for 5 h at room temperature. After the volatile materials were removed in vacuo, the resultant residue was washed with *n*-hexane (5 mL) to afford **2b** as a white solid (62.2 mg, 64%). ^1H NMR (300 MHz, $CDCl_3$): $\delta =$ 0.82 (s, 9H), 1.02 (s, 9H), 1.26 (s, 9H), 1.36 (s, 9H), 1.40 (s, 9H), 1.62 (s, 9H), 6.05–6.09 (m, 2H), 6.65–6.71 (m, 2H), 6.88 (t, 1H, $J =$ 2.9 Hz), 7.05–7.08 (m, 2H), 7.20–7.30 (m, 3H), 7.32–7.41 ppm (m, 4H); $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$): $\delta = 31.1$, 31.4, 32.2 (d, $J = 6.0$ Hz), 32.9 (d, $J = 6.7$ Hz), 33.6 (d, $J = 2.9$ Hz), 33.9, 34.7, 34.9, 37.2, 38.1, 39.8, 41.4, 123.1 (d, $J = 11.2$ Hz), 123.7 (d, $J = 7.7$ Hz), 126.3 (d, $J =$ 4.2 Hz), 126.7 (d, $J = 1.7$ Hz), 126.8 (d, $J = 6.3$ Hz), 127.5 (d, $J =$ 3.1 Hz), 127.9–128.3, 131.0 (dd, $J = 62.3$ Hz, $J = 5.9$ Hz), 132.5 (d, $J =$ 2.1 Hz), 137.2 (d, $J = 60.6$ Hz), 138.9 (dd, $J = 17.3$ Hz, $J = 4.5$ Hz), 151.1, 152.1 (d, $J = 2.6$ Hz), 154.6, 156.0 (d, $J = 3.7$ Hz), 158.9 (d, $J =$ 8.0 Hz), 160.7 (d, $J = 3.8$ Hz), 186.9 ppm (d, $J = 13.7$ Hz); $^{31}P\{^1H\}$ NMR (121 MHz, $CDCl_3$): $\delta = 107.6$ (d, $J = 124.4$ Hz), 188.9 ppm (d, $J =$ 124.4 Hz); HRMS (APCI) calcd for $C_{49}H_{68}AuClNaOP_2$ $[M+Na]^+$: 989.3997; found: 989.3976.

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Preparation of 3a

A mixture of **1a** (71.5 mg, 0.10 mmol) and (tht)AuCl (64.1 mg, 0.20 mmol) in dichloromethane (1 mL) was stirred for 1 h at room temperature. After the volatile materials were removed under reduced pressure, the resultant residue was dissolved in dichloromethane (1 mL) under nitrogen atmosphere. The mixture was stirred for 1 h at room temperature, and the volatile materials were removed under reduced pressure. The resultant residue was washed with *n*-hexane (5 mL) to afford **3a** as a pink solid (98.2 mg, 80%). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 0.85–0.91 (m, 1H), 1.03 (t, 3H, J = 7.3 Hz), 1.31 (s, 18H), 1.37 (s, 9H), 1.54 (s, 9H), 1.68 (s, 9H), 1.83 (s, 9H), 1.83–2.00 (m, 2H), 2.10–2.30 (m, 1H), 2.73–2.85 (m, 1H), 3.55–3.69 (m, 1H), 5.79–5.83 (m, 2H), 6.67 (pt, 2H, J = 7.2 Hz), 6.99–7.06 (m, 1H), 7.29–7.31 (m, 2H), 7.53–7.58 ppm (m, 2H); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$): δ = 14.1, 23.8 (d, J = 21.1 Hz), 31.1, 32.3 (d, J = 5.6 Hz), 33.9, 34.2, 34.3 (d, J = 1.7 Hz), 34.7, 35.4, 38.2 (d, J = 25.6 Hz), 38.9 (d, J = 2.0 Hz), 39.0 (d, J = 2.1 Hz), 40.5 (d, J = 3.9 Hz), 41.0, 118.7 (d, J = 32.3 Hz), 123.1 (d, J = 10.1 Hz), 123.9 (d, J = 7.7 Hz), 124.5 (d, J = 36.3 Hz), 125.3 (d, J = 9.5 Hz), 125.9 (d, J = 9.3 Hz), 127.1 (d, J = 10.1 Hz), 128.0 (d, J = 3.7 Hz), 130.7 (d, J = 5.8 Hz), 136.1 (d, J = 4.4 Hz), 153.1 (d, J = 4.5 Hz), 155.6 (d, J = 3.0 Hz), 157.4, 158.2 (d, J = 3.3 Hz), 158.6, 159.6 (d, J = 18.3 Hz), 183.5 ppm (dd, J = 103.2 Hz, J = 6.1 Hz); ^{31}P{^{1}H} NMR (121 MHz, $CDCl_3$): δ = 138.7 (d, J = 40.4 Hz), 154.0 ppm (d, J = 40.4 Hz); HRMS (APCI) calcd for $C_{47}H_{72}Au_2Cl_2NaOP_2$ $[M+Na]^+$: 1201.3664; found: 1201.3654.

X-ray crystallographic data of **3a**: $C_{47}H_{72}Cl_2OP_2Au2$ M_W = 1179.81, crystal dimensions = 0.160 × 0.160 × 0.080 mm^3, triclinic, $P\bar{1}$ (#2), a = 15.9562(5), b = 18.3089(6), c = 20.0597(8) Å, α = 116.9745(9), β = 98.5272(10), γ = 101.1501(5)°, V = 4934.0(3) Å^3, Z = 4, T = 143 K, λ = 0.71075 Å, ρ_{calc} = 1.588 g m^{-3}, $\mu_{MoK\alpha}$ = 6.145 mm^{-1}, F_{000} = 2336, $-20 \le h \le 20$, $-23 \le k \le 23$, $-26 \le l \le 25$, 79631 total reflections ($2\theta_{max}$ = 54.97°), 22476 unique (R_{int} = 0.0634), R = 0.0583 ($\sigma > 2\sigma(I)$), 0.1073 (all data), $wR2$ = 0.1477 ($\sigma > 2\sigma(I)$), 0.2350 (all data), S = 1.436 (898 parameters). See the Supporting Information.

Preparation of 3b

A mixture of **1a** (73.5 mg, 0.10 mmol) and (tht)AuCl (64.1 mg, 0.20 mmol) in dichloromethane (1 mL) was stirred for 1 h at room temperature. After the volatile materials were removed under reduced pressure, the resultant residue was dissolved in dichloromethane (1 mL) under nitrogen atmosphere. The mixture was stirred for 1 h at room temperature, and the volatile materials were removed under reduced pressure. The resultant residue was washed with *n*-hexane (5 mL) to afford **3b** as a pink solid (102.1 mg, 85%). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$): δ = 31.0 (2C), 33.8 (d, J = 3.5 Hz), 34.3, 34.5, 34.8, 35.1, 38.9 (d, J = 1.4 Hz), 39.9, 41.3, 122.8 (d, J = 10.4 Hz), 123.5, 124.8, 126.5–128.5, 129.4, 133.3, 135.0 (dd, J = 54.8 Hz, J = 0.7 Hz), 135.4–136.5, 153.1 (d, J = 3.0 Hz), 154.6 (d, J = 2.8 Hz), 157.5 (d, J = 1.4 Hz), 158.1, 159.3–159.7, 161.4, 190.6 ppm (d, J = 47.3 Hz); HRMS (APCI) calcd for $C_{49}H_{68}Au_2Cl_2NaOP_2$ $[M+Na]^+$: 1221.3351; found: 1221.3307.

Hydration of propargyl acetate[26]

To the corresponding chlorogold complex (3 mol%), a solution of propargyl acetate (17.4 mg, 0.10 mmol) in acetone (1 mL) and H_2O (1.0 mmol) was added at room temperature. After stirring for 48 h, the reaction mixture was directly loaded on a silica gel (SiO_2) column and eluted with hexane/ethyl acetate 6:1 to give 2-oxo-1-phenylpropyl acetate as a colorless oil.

Cycloisomerization of 1,6-enyne[27]

To the corresponding chlorogold complex (3 mol%), a solution of 1,6-enyne (25.3 mg, 0.10 mmol) in dichloromethane (1 mL) was added at room temperature. After stirring for 24 h, the reaction mixture was directly loaded on a silica gel column and eluted with hexane/ethyl acetate 6:1 to give the cycloisomerization product as a colorless oil.

Alkoxycyclization of 1,6-enyne[27]

To the corresponding chlorogold complex (3 mol%), a solution of 1,6-enyne (25.3 mg, 0.10 mmol) in dichloromethane (0.5 mL) and methanol (0.5 mL) was added at room temperature. After stirring for 24 h, the reaction mixture was directly loaded on a silica gel column and eluted with hexane/ethyl acetate 6:1 to give the product as a colorless oil.

Intramolecular hydroalkoxylation of allenyl alcohol[28]

A mixture of **3b** (5 mol%) and additive in toluene (0.4 mL) was stirred for 10 min at room temperature. To the mixture in toluene was added 2,2-diphenylhexa-4,5-dien-1-ol (25.3 mg, 0.10 mmol) in toluene (0.6 mL) at room temperature. After stirring for 24 h, the reaction mixture was directly loaded on a short silica gel column and eluted with hexane/ethyl acetate to give the 2-vinyltetrahydrofuran as a colorless oil.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: conformational isomerism · coordination chemistry · electron density distribution · gold catalysis · phosphaalkenes

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